Software Engineering Intern - Device Emulation

Technical Description

This position will require knowledge of APIs, network/web security, and custom CLI tools. You will be developing a CLI application in the language of your choice (Python, Java, Rust, C, etc) that takes a file path and transmits test data to our API endpoint. You will be implementing an industry standard web security protocol called mutual TLS. Your tool will be an “emulator” for the device that the hardware team is designing, and will be used substantially to test how the device will connect to our server.

We are using the Scrum (Agile) Development Methodology. You will complete work in two week sprints, presenting your plan at the beginning of the sprint and presenting the result of your work at the end.

Some potential challenges are that at least from the emulator/device side, there are some unknowns and you will have to do some independent research. It’s very important that if you don’t know the answer to something, you make the team aware so that we can plan for you to learn what you need. It’s understood and expected that you will not know everything - that’s okay.

During the sprint you will have access to a coach who is in charge of your work, and a mentor that is unrelated to your work but is there to support your personal and professional growth.

Project Overview

Our sponsor is the Adirondack Wilderness Advocates, a nonprofit organization that is devoted to keeping the wilderness areas in the Adirondack protected for future generations to enjoy. They have tasked us with assisting in their mission via a data driven platform that aggregates trail usage data for hikers, trail managers, and general administrators and presents it in a professional, easily understood web dashboard. There are several different use cases for the data, as a guide for hikers looking to understand trail conditions for future trips, as a record of usage and hotspots potentially in need of maintenance for managers, and as a platform for potential NYS DEC administration of trail usage, maintenance, and policy making.

The team has been split along lines of experience and interest, however these lines are not strict and are **often** crossed:

- UX/UI subteam: In charge of the look and feel of the website, as well as any web-based changes
- AWS subteam: In charge of backend functions, database configurations, and security policies.

There is a hardware component to this project that is being completed by a separate MSD Kate Gleason Engineering Team. This component is a device which records trail usage data and uploads it to our service via an API we expose.

What's Been Accomplished

It is easier to write about what has ***not*** been completed for this project. Currently, this project's MVP is close to completion, but a few major feature enhancements are still needed. These enhancements are those which the team is not experienced with or lack the knowledge to be confident in its swift and performant implementation.

To date, the following artifacts have been completed:

1. MVP requirements, design, flow, and use cases
2. MVP features
    a. Landing Page with view-able map
    b. User Login
    c. Trail data chart view
    d. Trail viewing filters
    e. Device association
    f. Import/Export of data
    g. Toggle-able views
3. Testing environments/infrastructure
    a. TST
    b. UAT
    c. PROD
4. CI/CD pipeline
5. Documentation
    a. Use
    b. Architecture
    c. Design
6. Major refactors
    a. UI
    b. Data model
    c. API
7. Security
    a. Oauth
    b. AWS roles

And after this is published, more features will have been completed. The tasks that will be completed during the summer semester are enhancements and MVP security features.

Preferred Candidate Attributes

We are looking for a driven and motivated candidate - we think your career development is important and we want someone who thinks the same. We will work hard to ensure that you learn everything you need to be dangerous in industry, but this requires that you be ready to learn.

We are also hoping that you have some experience, either via course work or demonstrable personal projects in web or network security, as well as with desktop application development.

Work Package

You will be responsible for the planning, design, and implementation of an MVP tool in the first half of the summer, as well as further enhancements for the remainder of the summer. It is hoped that this tool can be integrated with our CI/CD pipeline to ensure that at every release of our software, that the device is capable of communicating seamlessly with the server.